Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Soeren Becker	Email	soeren.becker@iap.fraunhofer.de
Institute Name	Fraunhofer Institute for Applied Polymer Research IAP-CAN			
Address	Grindelallee 117, 20146 Hamburg, Germany			

2. Technology Overview

Title of Proposed Technology	CANflow: Continuous Flow synthesis of quantum dot material with high quality for industrial production.			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum (√)			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Fine Chemicals	Nano-Application Technologies	Nano/Micro Mechanical Systems	Other Nano/Micro Mechanical System Technologies
Concept of Proposed Technology	The proposed technology, CANflow, is a microfluidic-based continuous-flow platform for the reproducible and cost-efficient production of high-quality quantum dot (QD) materials at industrially relevant throughputs. CANflow integrates micro-fluidic components to precisely control reaction conditions such as temperature, mixing, and residence time, enabling narrow size distributions, high quantum yields, and excellent batch-to-batch consistency. The technology has already been successfully demonstrated for established QD systems, including CdSe/CdS and PbS, and supports throughputs of up to 1 g of inorganic material per minute. This makes it suitable for scale-up towards industrial production while maintaining product quality and process stability. The resulting QDs are applicable in a variety of fields such as displays, lighting, and sensors, with a particular focus on display applications as the primary target market.			

(max 10 pages)

	Within this project, CANflow will be adapted and optimized for the next generation of quantum dot materials. The continuous-flow approach will enable scalable, flexible, and efficient manufacturing, providing a robust technological foundation for future commercial production by the industrial partner.
Rationale for Joint R&D Project	A joint R&D project is essential to bridge the gap between advanced continuous-flow synthesis technology and the industrial production needs for next-generation quantum dot (QD) materials. The industrial partner requires a reliable, scalable, and cost-effective technology platform to manufacture QDs at industrial scale, tailored to specific product and market requirements, particularly in the display sector. The CANflow technology provides a mature microfluidic-based continuous-flow process that can be flexibly adapted to different QD systems. Within the collaboration, CANflow can either be integrated with an existing QD synthesis route of the partner or used as a basis to co-develop new QD materials that meet the partner's performance and regulatory requirements. This co-development approach ensures that material properties (e.g., emission wavelength, stability, toxicity profile) and process characteristics (e.g., throughput, cost, robustness) are aligned with the partner's application and commercialization strategy. By combining the applicant's expertise in continuous-flow process engineering with the industrial partner's knowledge of market needs, product specifications, and downstream integration, the joint project will accelerate the transition from laboratory-scale innovation to industrial-scale manufacturing. This collaborative framework reduces technical risk, shortens development cycles, and increases the likelihood of successful commercialization of next-generation QD materials.
Key Words	1. quantum dots 2. micro-fluidic synthesis 3. emitter for displays and lighting 4. absorber material for sensors 5. colloidal nanoparticles

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (5)	End of the Project	Level (8)
R&D Objectives	1. **Adaptation of the CANflow reactor to novel quantum dot materials** Develop and optimize reactor configurations and process parameters that enable the continuous-flow synthesis of next-generation quantum dot (QD) materials at industrially relevant scale, while maintaining high product quality and reproducibility. 2. **Transfer of existing quantum dot synthesis protocols to the CANflow platform** Translate and refine established batch or semi-batch QD synthesis routes from the industrial partner into continuous-flow processes using CANflow, including optimization of reaction conditions to achieve targeted optical and structural properties. *or* **Development of new quantum dot materials using the CANflow reactor** Co-develop innovative QD compositions and formulations directly within the CANflow system, tailored to the partner's application requirements (e.g., for displays, lighting, sensors), focusing on improved performance, stability, and regulatory compliance. *3.* **Scale-up and process robustness for industrial implementation** Establish scalable process windows, in-line monitoring strategies, and quality-control protocols to ensure stable, high-throughput production of QDs and to prepare the technology for seamless transfer to industrial manufacturing environments.			
R&D Target Specifications	• **Production scale:** ○ Upscaling of quantum dot (QD) synthesis to a (multi-)kilogram annual production capacity under continuous-flow conditions. • **Optical performance:** ○ Quantum yield (QY) of **> 90%** for the targeted QD materials. ○ Narrow emission bandwidth (FWHM) appropriate for display applications (e.g., ≤ 30 nm, if technically feasible for the selected material system). • **Product quality and stability:** ○ Narrow particle size distribution with standard deviation ≤ 10% of the mean particle diameter. ○ Chemical and photostability compatible with typical display operating conditions (e.g., negligible degradation over standardized accelerated aging tests). • **Process performance:** ○ Stable, continuous operation with minimal deviation in key process parameters (temperature, flow rate, residence time) during long-term runs (e.g., ≥ 8 h). ○ Reproducibility of QD properties within a defined tolerance between runs (e.g., variation in QY and peak emission wavelength ≤ 5%).			
R&D Plan	The R&D project is structured into three main work packages (WPs) that build sequentially on each other, from material and protocol selection to industrially relevant scale-up. Each WP has clearly defined objectives and timelines, with a critical milestone after 12 months to demonstrate the feasibility of the CANflow-based quantum dot production. WP1 (3 months): Selection and evaluation of quantum dot material and the synthesis protocol WP2 (9 months): Transfer of quantum dot synthesis to CANflow reactor (gram scale) WP3 (12 months): Up-scaling of quantum dot flow synthesis to kilogram scale Milestone after 12 months: successful transfer of synthesis to CANflow reactor (critical)			

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute is expected to lead the technical development and implementation of the CANflow-based quantum dot (QD) production processes. Its primary role will be the transfer of existing batch QD synthesis protocols to the CANflow reactor, including the adaptation and optimization of reaction parameters for continuous-flow operation. The institute will be responsible for comprehensive characterization of the resulting QDs (e.g., optical, structural, and stability analysis) to ensure that all target specifications are met. In addition, it will design the layout and perform the set-up of the up-scaled flow synthesis system, defining reactor configurations and process control strategies suitable for higher throughputs. Finally, the research institute will demonstrate large-scale QD production using the CANflow technology, validating process robustness, reproducibility, and performance at (multi-)kilogram per year scale.
Expected R&R of Korean SME Partner	The Korean SME partner is expected to define the target quantum dot (QD) systems based on its application and market requirements, particularly for display and related fields. Where available, the SME will provide existing batch synthesis protocols and relevant know-how as a basis for their transfer to the CANflow reactor. Furthermore, the SME will be responsible for evaluating the performance of the QDs produced via CANflow, using application-relevant test procedures (e.g., device integration tests, reliability and lifetime measurements). This feedback will guide further optimization of the synthesis and ensure that the developed materials meet the SME's product specifications and commercialization needs.

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint R&D project is expected to deliver several key outcomes in terms of technology, commercialization, and broader industrial impact. Technologically, the main objective is to develop and demonstrate a robust, continuous-flow synthesis process for quantum dots (QDs) at industrial scale. The project will build on the applicant's existing flow technology and adapt it to enable highly controlled, reproducible, and scalable QD production. This includes optimizing process parameters to achieve consistent particle size distribution, high quantum yield, and long-term stability of the QDs. By the end of the project, the process shall be ready for transfer to the industrial partner, who will use it for commercial-scale QD manufacturing. From a commercialization perspective, the project will significantly reduce the gap between laboratory-scale synthesis and high-volume industrial production. The industrial partner will be in a position to produce QDs with stable quality, high throughput, and improved cost-efficiency, enabling a competitive product portfolio in target application areas. The primary target market is the display industry, where high-quality QDs are used for advanced display technologies with improved color gamut, brightness, and energy efficiency. Additional application fields include lighting, sensor technologies, and potentially other optoelectronic components. The CANflow technology is expected to lower production costs, enhance scalability, and shorten time-to-market, thereby improving the overall business case for QD-based products. On a broader industrial and economic level, the project will support the establishment of a reliable value chain for quantum dot materials. The industrial-scale continuous-flow process can serve as a platform technology that can be adapted to other nanomaterials, further amplifying its industrial relevance. The project will also create or secure high-skilled jobs in R&D, process engineering, and production. Overall, the outcomes are expected to accelerate the adoption of QD-enhanced display technologies, stimulate downstream innovation in electronics and sensor markets, and contribute to sustainable economic growth through advanced manufacturing capabilities.

6. References

(1) KR101369691B1

(2) KR102141505B1

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization